

Table of Contents

1.0 Introduction	3
2.0 Literature Review	4
3.0 Design	5
4.0 Design Analysis	7
Design Requirements	7
Design Concepts	8
Bearing Selection	9
Pneumatic Cylinder Selection	9
End Effectors	15
5.0 Safety Analysis	17
Safety Considerations and Features	17
OSHA Regulations	17
6.0 Logic Analysis	19
Pneumatics Analysis	19
Boolean Analysis	20
7.0 PLC Model	22
8.0 Conclusions and Recommendations	23
9.0 References	24
10.0 Appendix	25

1.0 Introduction

The following report details the motor house bearing project which looks into the installation of a bearing into a housing of an electric motor. This project implements a pneumatic system as well as a PLC to automate the bearing installation process, streamlining operations and ensuring precision.

The pneumatic system operates in a defined sequence, driven by inputs from proximity sensors that detect range and trigger the next stage of operation and limit switches on the cylinders which provide feedback to the system. These sensors ensure accurate alignment and efficient progression through the installation process. The control logic is implemented using ladder logic programming, which utilizes timers and condition-based triggers to manage the sequence.

By integrating these technologies, the project demonstrates how automation and sensor-driven feedback can enhance the reliability and efficiency of manufacturing processes. This report details the design and development of the pneumatic system and the ladder logic sequence, supported by theoretical analysis of stress/force calculations.

2.0 Literature Review

The following literature review concentrates on Pick and Place machines and bearing installation systems. The pick and place systems are commonly used within the electronics, automotive, food and beverage, and pharmaceuticals industries. Performing tasks such as packaging, sorting, and assembling. Whereas, bearing installation is commonly needed in the automotive and machinery industries, it is crucial to have precise bearing alignment for equipment performance as it corresponds to longevity and reliability of the product.

Cycle time and efficiency are critical factors in the performance of pick-and-place and bearing installation machines. Optimizing these parameters enhances productivity and reduces operational costs. Automating these processes ensures manufacturing at high speeds and accuracy. This technology performs repetitive tasks with high precision, freeing manual labor for more complex activities and boosting overall efficiency. For example, in automotive manufacturing, pick-and-place robots are used to assemble parts into car bodies, installing seats, and mounting tires. These robots can assemble up to 200 products per minute, their ability to handle heavy loads and perform repetitive tasks efficiently makes them ideal for high-volume production environments.

Common limitations that the pick and place machine faces is with the variation in product design and size, often requiring reprogramming or mechanical adjustments. For bearing installation issues commonly faced are with achieving exact alignment, since as discussed above, misalignment can lead to premature wear and failure. Another factor is force control, ensuring the appropriate amount of force is applied to press fit the bearing as insufficient force would lead to a loose fit, however, not applying excessive force causing damage to the components.

The trends seen within the industry to improve the system have adapted the usage of advanced sensors and AI. The use of vision systems, force sensors, and proximity sensors improves the accuracy and reliability of both pick-and-place and bearing installation machines. The integration of artificial intelligence (AI) and machine learning enables pick-and-place machines to handle complex tasks with higher precision and adaptability.

3.0 Design

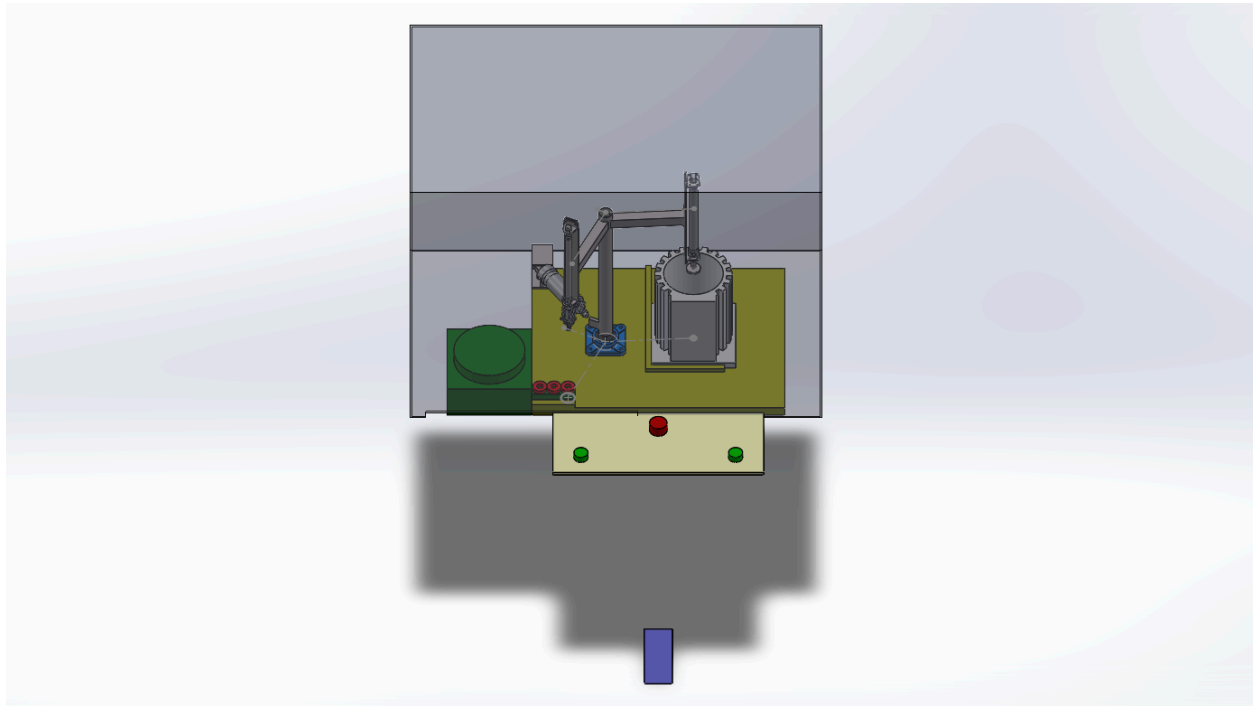


Figure 1: Final Assembly of the design

The main element of this design is the main shaft which is secured to the workstation using a Flange Bearing mount sourced from Gainer (item # WWG36UY36). This mount is bolted down to the workstation by 4 bolts with a diameter of $\frac{1}{2}$ ". Additionally, the main shaft, made of 4130 steel, is secured onto this mount with the use of two set screws that are provided with the mount as well. This shaft is a round tube of 4130 steel with an ID of 1.875", OD of 2" and length of 26". It contains two arms which are 118.66 degrees apart from each other, and each has a specific length, the press cylinder arm has a length of 11.418" from the centre of the circular tube, and the grip cylinder arm is 11.391". Also, both arms are placed 23.56" above the button of the shaft to ensure that each cylinder will be able to move with a clearance of 0.75" with respect to the electric motor housing. Each arm also has a $\frac{1}{8}$ " plate protruding at the end, which is where the cylinders are mounted. The mounting plate of the gripper arm is 17.25" x 2.25" x 0.125" and made from 4130 Steel and similarly the plate of the press arm is 16.75" x 2.75" x 0.125".

For the gripper cylinder, the DSNU-20-300-PPV-A cylinder along with the DHPC-10-A-S gripper from the Festo catalogue are used. The DSNU-20-300-PPV-A cylinder has a piston diameter of 20mm and stroke length of 300mm. The DHPC-10-A-S gripper has a gripping force of 21.5N and stroke of 2 mm per jaw. Custom fingers, made of 4130 steel are designed to fit on the inside of the jaws so that when the gripper fingers extend, the fingers will clasp the inner race of the bearing. These fingers were given a radius of 0.437500 at the ends so that they would better grab the inner race of the bearings. The design of the fingers allow for a quick replacement if the fingers are to be damaged, because they will simply be threaded through the jaws via bolt and nut. Figures 8 to 10 show the FEA conducted on the main shaft when the cylinder extends down towards the bearing (as extension force is higher than retraction). From the figures, it can

be noted that the maximum stress on the shaft is $2.97 \times 10^7 \text{ N/mm}^2$, the maximum deflection is 1.269mm and a minimum factor of safety of 1.549. These values indicate that the main shaft is rigid enough to elastically deform from the force of the gripper arm extending, but it is able to handle 1.5 times that load before plastically deforming.

For the press cylinder, the DSNU-32-300-P-A-S10 cylinder is used from the Festo catalogue, as it has a diameter of 32mm, stroke length of 300mm and a theoretical extending force of 483 N, which is required to press the bearing into the motor housing with a 100 lbf (444.8N) of force. Figures 11 to 13 show the analysis done on the main shaft when the press cylinder is to extend as full force. From the studies, it can be seen that the maximum stress would be $7.693 \times 10^7 \text{ N/mm}^2$, the maximum deflection would be 2.964 mm and the minimum factor of safety would be 5.979. Similar to the analysis on the main shaft from the gripper cylinder, it can be seen that these values indicate the design is strong enough to withstand any forces it may experience.

A custom stamper part has been designed to thread onto the end of the cylinder, and its design allows it to press the bearing into the motor housing by only pushing on its outer race. The stamper is made of 4130 steel with a shaft OD of 16mm and ID of 10 mm. At the end of this shaft is a larger circular shape with an OD of 2" and ID of 1.763780". A fillet with a radius of 0.25" is applied to the connection between the shaft and the shape to add more strength to the design. From the analysis shown in Figures 14 to 16, it can be seen that with the force of 443N applied downwards, and the force of 110 lbf (444.8 N) applied in the opposite direction, the stamper has a maximum stress of $1.27 \times 10^7 \text{ N/mm}^2$, a maximum displacement of $3.073 \times 10^{-3} \text{ mm}$ and a minimum factor of safety of 3.6, hence showing that it is strong enough to withstand the forces applied to it.

Finally, safety features such as an enclosure, two hand push buttons, foot pedal, and emergency stop are included to ensure that operator safety is thought about as well as to meet the project requirements of the buttons and pedal.

4.0 Design Analysis

Design Requirements

1. The machine must be bolted on the pre-existing workstation table.
 - a. Dimensions of the workstation are shown in Figure 1.
2. The machine must pick up the bearings from the bearing feeder.
 - a. Machine must use the inner race of the bearings to pick them up.
3. The machine must place the bearings into the motor housing.
4. The machine must press fit a bearing into a motor housing.
 - a. A force of 100 lbf is required to press the bearing.
 - b. Cycle time must be less than 30 seconds.
5. The machine must apply force to only the outer race.
 - a. Bearing dimensions are as follows, ID=0.8750" OD=2.000" Height=0.6250".
6. The machine must use pneumatics only for motion.
 - a. Pneumatics must be selected from the Festo Catalog - DSNU line of cylinder.
 - b. Maximum supply air pressure is 90 psi.
7. The machine must use a PLC control system.
 - a. The PLC has to be an Allen-Bradley Micrologix 1400.
 - b. PLC sensors have to be 24V and compatible with the above PLC.
8. The machine must keep the worker safe
 - a. The machine must meet with all OSHA regulations.

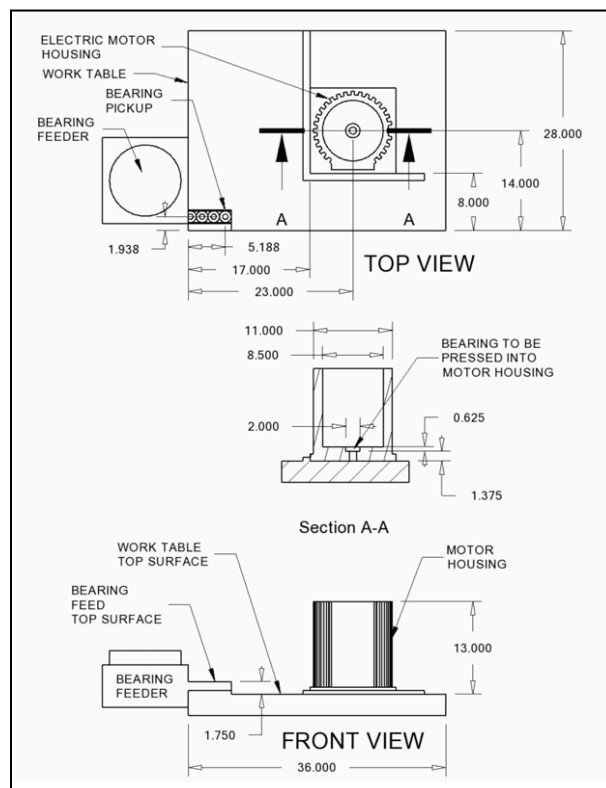


Figure 2. Existing Bearing Assembly Workstation

Design Concepts

Several pneumatic cylinders are needed for this machine. One pneumatic cylinder is required to pick up the bearings from the bearing feeder and place them into the motor housing, one cylinder is required to press fit the bearings, and one or more cylinders are required for the movement of other cylinders. The cylinder that will pick up and move the bearings will have to be fitted with a gripper. The gripper cylinder's piston will extend downwards so that the gripper can grasp the bearing by its inner race then retract and move to the motor housing. The piston will then extend to place the bearing, the gripper will release the bearing, and the piston will retract and move back to the bearing feeder. The press fit cylinder will extend to press the bearing into the motor housing then retract. The press fit cylinder must also have an end effector that only applies force to the outer race of the bearing. The cylinder(s) that will provide the motion can move the other cylinder(s) either linearly or angularly. It is important to consider the transverse force limit of the cylinders to ensure they are capable of supporting required weight. The pneumatics will be controlled using an Allen-Bradley Micrologix 1400 PLC.

Due to dimensional constraints having a rotating assembly be more effective than a linearly moving assembly. The cylinder that will produce this motion will be CYL-Move which will rotate an assembly holding the other two cylinders, the gripper cylinder (CYL-Grip) and the press fit cylinder (CYL-Press). The assembly may be a simple shaft mounted to the workstation using a thrust bearing with extruding arms with the cylinders mounted to the ends. If the assembly's point of rotation is an equal distance from the bearings pick up position and the bearing placement position, and CYL-Grip and CYL-Press are mounted the same distance from the point of rotation, CYL-Move would only need to extend and retract once to produce the required movement. For CYL-Move to move the shaft angularly, the joints need to have some degree of freedom to allow for the cylinder to fully extend. To do this the cylinder can be mounted to a rotating joint and the piston rod end can also be connected to a rotating joint mounted to the shaft.

CYL-Grip and CYL-Press can be mounted to arms extruding from the main shaft. This will require mounting plates as well so there is a surface to bolt the cylinders to. Mounting accessories can be chosen from the Festo catalogue. There may also need to be an arm lower on the main shaft for CYL-Move to connect to. CYL-Grip requires a gripper capable of picking up the bearing; this gripper may be pneumatic and also chosen from the Festo catalogue. CYL-Press requires an end effector capable of applying force to only the outer race of the bearing; this may need to be a custom part. The dimensions of the assembly and the layout will be based on the required stroke length, the amount of space available, and the size of the cylinders and accessories.

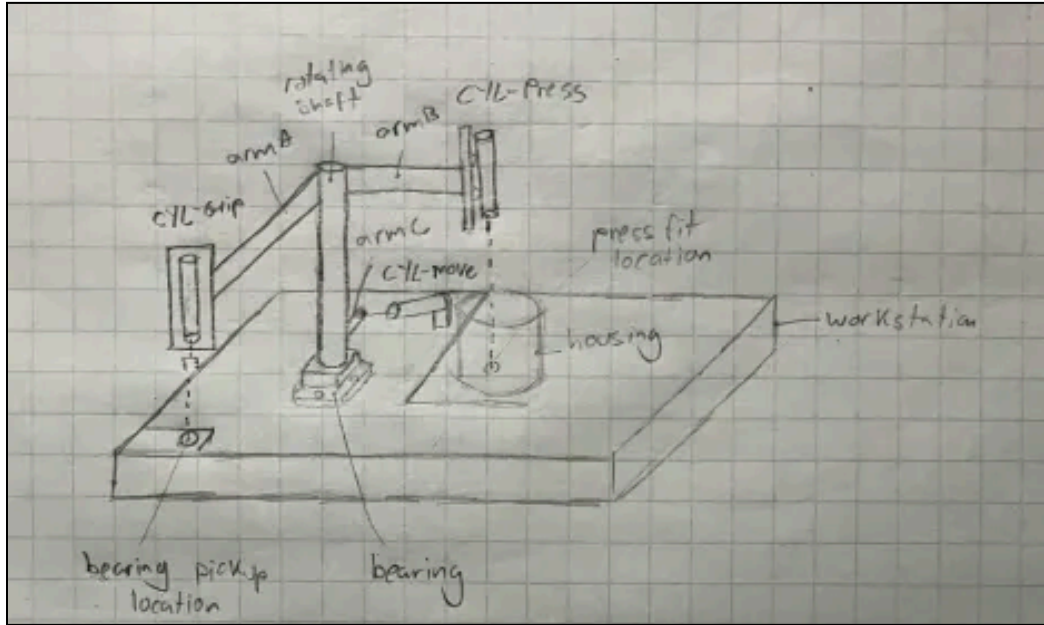


Figure 3. Schematic of design

Bearing Selection

An off the shelf thrust bearing was chosen to support the shaft and connect the assembly to the workstation [8]. The main shaft is planned to have an outer diameter of 2" therefore a bearing was chosen with an inner diameter of 2". This bearing allows the shaft to sit flush with the table therefore the shafts height will be relative to the workstation. The bearing also has a height of 2.15" so all features must be connected above this. The bearing can be bolted directly to the surface of the workstation.

Pneumatic Cylinder Selection

The pneumatic cylinders will be selected based on the required force, the required stroke length, the weight they must support, and how much space is available. The cylinders will be chosen from the Festo Catalogue's DSNU line of cylinders.

Gripper Cylinder (CYL-A)

The required force CYL-A must exert depends on the weight of the gripper and the bearing. The gripper that was chosen has a mass of 110g [1]. The mass of a basic ball bearing that's 2" in diameter is calculated below as well as the force required from the cylinder (assuming no acceleration):

$$\rho_{steel} \approx 0.28 \text{ lbm/in}^3 [9]$$

$$V_{bearing} \approx \frac{\pi}{4} (OD^2 - ID^2)h = \frac{\pi}{4} (2^2 - 0.875^2)(0.625) = 1.59 \text{ in}^3$$

$$m_{bearing} = \rho_{steel} V_{bearing} = (0.28 \text{ lbm/in}^3)(1.59 \text{ in}^3) = 0.445 \text{ lbm} = 202 \text{ g}$$

$$m_{gripper} = 110g$$

$$\Sigma F = ma = 0 = F_{CYL-A} - Fg$$

$$F_{CYL-A} = m_{gripper+bearing}g$$

$$F_{CYL-A} = (0.11 + 0.202kg)(9.81)$$

$$F_{CYL-A} = 3.06N = 0.688lbf$$

Since the required force and the maximum operating pressure is known (90psi), the minimum piston diameter can be calculated using the formula for pressure.

$$P = F/A$$

$$P = \frac{4F}{\pi d^2}$$

$$d = \sqrt{\frac{4F}{\pi P}}$$

$$d = \sqrt{\frac{4(0.688lbf)}{\pi(90psi)}}$$

$$d = 0.0987" = 2.5mm$$

The smallest piston size is 8mm diameter therefore this piston size should be able to support the weight of the load, however, the stroke length of the 8mm pistons are not long enough for the requirements therefore a 20mm diameter piston is required.

The minimum stroke length of CYL-A is the distance the cylinder must extend to place the bearing in the motor housing from the top of the housing. Bearings are placed in the bottom of the housing, which is 11" from the top of the housing; therefore CYL-A requires a stroke length that is greater than 11" (279.4mm). The stroke must be greater than the actual distance to provide some clearance for the cylinder to move above the housing from one position to the next.

Table 1. Gripper Cylinder requirements

Piston Diameter	Stroke length
$\geq 2.5mm$	$> 279.4mm$

The compact version (DSNU - S) of the cylinder does not meet the stroke length requirement therefore a DSNU - Series - Round double-acting cylinder will be chosen. The maximum operating pressure must be 90psi; this series has maximum operating pressure of 145 psi. The closest stroke length available to the required stroke is 300mm which provides about 20 mm of clearance when moving above the housing. This cylinder should travel at a high speed to ensure the cycle time is as short as possible therefore standard motion will suffice. The cushioning type for this cylinder should be PPV due to the high speeds it will travel at. This cylinder will be mounted vertically to the assembly with a foot mounting.

Table 2. Selection for Gripper Cylinder (CYL-A)

Series	Round cylinder, double-acting, based on ISO 6432	
Max. Operating pressure	1MPa (~145psi)	
Piston Diameter	20mm	
Stroke Length	300mm	
Cushioning Type (P/PPV/PPS)	PPV	
Constant Motion (standard / uniform, slow)	Standard	
Type Code	DSNU-20-300-PPV-A	
Part #	19244	

Table 3. Accessories - CYL-Grip

Cylinder: CYL-A	
Mounting	Piston Rod End
Foot mounting HBN/CRHBN	Parallel Gripper
Part #: 5127	Part #: 8116788

Press Fit Cylinder (CYL-B)

CYL-B must be able to exert a force of 100 lbf (444.8N) at an operating pressure of 90 psi to press fit the bearing. Since the force required and the pressure is known the piston diameter can be calculated using the formula for pressure.

$$\begin{aligned}
 P &= F/A \\
 P &= \frac{4F}{\pi d^2} \\
 d &= \sqrt{\frac{4F}{\pi P}} \\
 d &= \sqrt{\frac{4(100\text{lbf})}{\pi(90\text{psi})}} \\
 d &= 1.189" = 30.20\text{mm}
 \end{aligned}$$

Therefore the piston diameter should be at least 30.2mm to meet the force requirement. The stroke length of the cylinder should be long enough so that the piston will reach the bearing at the bottom of the motor housing when fully extended and be completely out of the motor housing when fully retracted, otherwise an extra cylinder would be required to raise and lower this cylinder. The distance to the bearing from the top of the motor housing is 11", thus the stroke length should be greater than 11" (279.4mm) so the cylinder has some clearance above the housing.

Table 4. Press Fit Cylinder requirements

Piston Diameter	Stroke length
$\geq 30.20\text{mm}$	$> 279.4\text{mm}$

The compact version (DSNU - S) of the cylinder does not meet the force requirement therefore a DSNU - Series - Round double-acting cylinder will be chosen. The maximum operating pressure must be 90psi; this series has maximum operating pressure of 145 psi. The closest stroke length available to the required stroke is 300mm which provides about 20 mm of clearance when moving above the housing. The motion should be relatively slow to prevent damaging the bearings during the press fit operation, therefore, the motion type should be uniform and slow. This cylinder type allows for speeds between 8 and 100mm/s with no load at operating pressure of 0.6MPa for a 32mm piston. Since this cylinder's motion is relatively slow and does not lift any loads, the P type cushioning should suffice. This type of cushioning consists of elastic cushioning rings on both sides of the cylinder and is for low speeds and low impact energies [1]. This cylinder will be mounted vertically to the rotating assembly with a Foot mounting.

Table 4. Selection for Press fit Cylinder (CYL-B)

Series	DSNU - Round cylinder, double-acting, based on ISO 6432	
Max. Operating pressure	1MPa (~145 psi)	
Piston Diameter	32mm	
Stroke Length	300mm	
Cushioning Type (P/PPV/PPS)	P	
Constant Motion (standard / uniform, slow)	Uniform, slow	
Type Code	DSNU-32-300-P-A-S10	
Part #	5249372	

Table 5. Accessories - CYL-Press

Cylinder: CYL-B	
Mounting	Piston Rod End
Foot mounting HBN/CRH	Custom Press attachment
Part #: 195851	

Cylinder for Movement (CYL-C)

The stroke length of CYL-C depends on the amount of rotation required as well as the length of the moment arm. The piston must rotate the assembly 118.7° ; a longer moment arm will reduce the force required, the moment arm also cannot be too short or the cylinder will not be able to complete the rotation around the shaft. Using the chord length formula the length of the moment arm can be calculated based on the angle of rotation and the stroke length. For a moment arm of 2" the chord length (stroke length) is 87mm. Since this length is not standard, a 100 mm stroke length will be used and the moment arm length can be calculated as follows:

$$\text{Stroke length} = L_{\text{chord}} = 2r\sin\theta$$
$$r = \frac{\text{Stroke length}}{2\sin\theta} = \frac{100\text{mm}}{2\sin(118.7^\circ)} = 2.288"$$

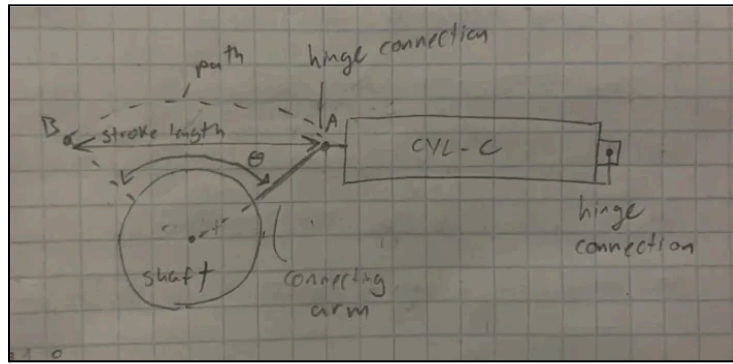


Figure 4. Schematic of rotating mechanism

CYL-C must exert the required force on the extend as well as the retract stroke to be able to move the assembly. The required force is dependent on the moment arm, how fast the assembly must accelerate, and the moment of inertia of the assembly. There are some resistive forces present however because of the bearing these forces can be neglected. To meet the cycle time requirement the cylinder should reach its maximum speed within about 2sec. The approximate maximum speed was taken from the catalog as 1m/s [1]. Assuming the cylinder is accelerating during the entire stroke length, and the assembly must rotate 118.7° in about 2 seconds, then the angular speed and acceleration can be calculated as follows:

$$\omega_f = \frac{v_{\text{max}}}{r} = \frac{1\text{m/s}}{2"} = \frac{1\text{m/s}}{0.0508\text{m}} = 19.69\text{rad/s}$$
$$\alpha = \frac{\Delta\omega}{\Delta t} = \frac{\omega_f - \omega_i}{\Delta t} = \frac{19.69\text{rad/s}}{2\text{s}} = 9.845\text{rad/s}^2$$

Since the angle between the cylinder and the moment arm changes as it rotates, the required force from the cylinder constantly changes. The force required from the cylinder is the greatest at the start of the retraction stroke. At this point the cylinder is at an angle of about 30° , therefore the cylinder must exert double the required force to rotate the cylinder at the required acceleration:

$$F_x = F\sin\theta = F\sin 30 = 1/2 F$$

The mass moment of inertia of the rotating assembly was taken from Solidworks and is $I_{xx} = 3.58 \text{ kg/m}^2$. The torque formula can be used to determine how much force is required from the cylinder based on the moment arm.

$$\begin{aligned}\Sigma F_{\perp} r &= F_{CYL} r \\ \Sigma M_o &= \Sigma F_{\perp} r = I_o \alpha \\ F_{\perp} &= \frac{I_{xx} \alpha}{r} = \frac{(3.58 \text{ kg/m}^2)(9.845 \text{ rad/s}^2)}{0.0508 \text{ m}} = 694 \text{ N} \\ F_{CYL} &= 2F_{\perp} = 1388 \text{ N}\end{aligned}$$

Based on the calculation a force of about 1388N of is required to rotate the assembly within 2 seconds with a moment arm length of 2.288". The moment arm length is taken from the centre of the main shaft to the connection of the piston rod end. Since the main shaft has a radius of 1", the arm connecting the cylinder to the shaft will be 1.288" long. Furthermore, since the required force on the retraction stroke and the operating pressure is known the minimum piston diameter can be calculated. The required piston diameter for the extension stroke will be calculated first then based on the diameters available, the area for the retract stroke will be checked with the force requirement.

$$\begin{aligned}P &= F/A \\ P &= \frac{4F}{\pi d^2} \\ d &= \sqrt{\frac{4F}{\pi P}} \\ d &= \sqrt{\frac{4(1388 \text{ N})}{\pi(0.621 \text{ MPa})}} \\ d &= 53.3 \text{ mm}\end{aligned}$$

The closest diameter piston to 53.3mm that also meets the force requirements is the 63mm piston. The information provided in the catalog states that the 63mm piston can exert 1682 N of force [1] during the retraction stroke at a lower operating pressure than what will be used for this design, therefore the 63mm piston will meet the force requirements.

Table 6. Movement Cylinder requirements

Piston Diameter	Stroke length
63mm	100mm

The compact version (DSNU - S) of the cylinder does not meet the stroke length requirement therefore a DSNU - Series - Round double-acting cylinder will be chosen. The maximum operating pressure must be 90psi; this series has maximum operating pressure of 145 psi. This cylinder should travel at a high speed to ensure the cycle time is as short as possible therefore standard motion will suffice; this cylinder will also carry a medium to heavy load. The cushioning type for this cylinder should be PPV due to the high speeds it will travel at. This cylinder will be mounted horizontally to the workstation table with a clevis foot mount and a rod clevis will be connected to the piston rod end.

Table 7. Selection for Movement Cylinder (CYL-C)

Series	Round cylinder, double-acting, based on ISO 6432	
Max. Operating pressure	1MPa (~145 psi)	
Piston Diameter	63mm	
Stroke Length	100mm	
Cushioning Type (P/PPV/PPS)	PPV	
Constant Motion (standard / uniform, slow)	Standard	
Type Code	DSNU-63-100-PPV-A	
Part #	196054	

Table 8. Accessories - CYL-move

Cylinder: CYL-C	
Mounting	Piston Rod End
Clevis foot LBN/CRLBN	Rod Clevis
Part #: 195862	Part #: 6146

End Effectors

Gripper

The gripper chosen is an off the shelf parallel double acting pneumatic gripper from the Festo catalogue and will be mounted to the cylinder's end rod. Custom gripper fingers are also required and will include a rubber covering to increase the grip force. The distance between the gripper finger mounts should be at least the inner diameter of the bearing, which is 0.875", or about 22.22mm. From the Festo catalogue, the smallest parallel gripper available that meets this requirement has a piston diameter of 16mm with the distance between the finger mounts being 25.8mm. This size should work since the fingers will take up some space as well. The required grip force can be calculated using the mass of the bearing which was calculated previously.

$$m_{bearing} = 0.445lbm$$

$$\mu_{rubber\ on\ steel} = 0.86 [10]$$

$$\Sigma F_y = 0 = mg - \mu F_{grripper}$$

$$F_{grripper} = \frac{mg}{2\mu} = \frac{(0.445lbm)(32.2)}{2(0.86)} = 8.33lbf = 37.1N$$

The catalog provides graphs for the grip force at various operating pressures and distances from the mounts. For the 16mm piston operating at 6 bar the grip force is between 60 and 65N with a maximum distance from the mounts being 30mm. Since this operating pressure is below the operating pressure that will be used, this gripper meets the force requirements. This gripper can be mounted from underneath directly to the end rod of CYL-Grip and has a weight of 110g. The gripper also requires sensors to control the movement. From the catalog a magneto-resistive proximity switch can be attached to the gripper via the C-slot on the side. This switch will be used to signal when CYL-Grip is fully extended by detecting the surface it is approaching. The gripper finger will also be designed to hold a contact sensor so that they can detect when the gripper is in contact with the bearing.

Table 9. Accessories - Gripper

Part	Part #
Parallel Gripper	8116788
24 V Proximity Sensor	547862

Press fit

CYL-Press should have an end effector that applies force to the outer race only, which has a diameter of 2". The end effector can be designed as a simple thin walled hollow cylinder with an outer diameter of 2" and a wall thickness of 0.24". To ensure the bearing can withstand the pressure the stress it can withstand (yield strength) will have to be compared to the applied stress. The press fit cylinder has a piston diameter of 32mm (1.26"); the force it produces must be calculated to determine the applied stress.

$$\sigma_{yield_steel} = 325 \text{ MPa} [9]$$

$$F_{applied} = PA = P \frac{\pi d^2}{4} = 90 \text{ psi} \frac{\pi (1.26")^2}{4} = 112.2 \text{ lbf}$$

$$\sigma_{applied} = \frac{F_{applied}}{A} = \frac{4F_{applied}}{\pi(OD^2 - ID^2)} = \frac{4(112.2 \text{ lbf})}{\pi(2^2 - 1.76^2)} = 158 \text{ psi} = 1.09 \text{ MPa}$$

The applied stress is significantly less than the yield strength for steel therefore the bearing will be able to withstand the applied pressure from the cylinder and the end effector.

5.0 Safety Analysis

Safety Considerations and Features

- 1) Enclosures- Address the issue of pinch points:
The system has moving parts (i.e cylinder, gripper), these parts must be covered to prevent accidental contact while the machine operates. The enclosure also must be clear, in order, to have full visibility of the machine operating. Additionally, warning labels will be placed near all pinch points to alert workers of the possibility of danger.
- 2) Interlocks- Address the issue of safe initiation of cycle:
The machine will not start unless the enclosures are closed as a measure of safety interlocking. Sensors will detect whether there is a motor housing in the work area before starting the cycle. The system will only start when both push buttons and the foot pedal are pressed simultaneously. Lastly, there will be a key system to lock out the machine during maintenance and to prevent unauthorized usage.
- 3) Emergency Stop Button- Address immediate stop of the machine:
When the button is pressed the system will immediately depressurize the pneumatic components (i.e cylinders) and stop all motion. The press-fit cylinder will be in retract mode to prevent damage to the motor housing. The other cylinders will default to the fail-safe mechanism and retract as well. The same procedure will occur if there is power loss or air loss.
- 4) Mufflers- Address possible health complications due to hazardous exposure to worker
Mufflers dampen the high-frequency noise produced by rapid air exhaust, reducing overall noise levels near the machine.

OSHA Regulations

The following table lists out the OSHA Regulations and Mandates required for the system:

Table 11: List of OSHA Requirements for Motor Housing Bearing Installer

Regulation	Purpose	Description
29 CFR 1910.212	General Requirements for Machine Guarding	• Machines must have guarding to protect operators and other employees from hazards such as moving parts, pinch points, and flying debris. • Guards must prevent hands, arms, or other body parts from reaching into the danger zone during operation. • Fixed, interlocked, or adjustable guards are required for all hazardous areas.
29 CFR 1910.147	Control of Hazardous Energy- Lockout/Tagout	• All energy sources (electrical, pneumatic, hydraulic) must be isolated and locked out during maintenance or servicing.

		● Machines must have a documented LOTO procedure. ● Operators and maintenance personnel must be trained in LOTO procedures. ● LOTO devices, such as keyed switches or padlocks, must prevent accidental reactivation of the machine.
29 CFR 1910.147	Control of Hazardous Energy- Emergency Stop Requirements	● Emergency stop devices must be accessible, easily recognizable, and independent of the machine's control logic. ● Emergency stops must immediately disconnect hazardous energy and remain locked until manually reset.
29 CFR 1910.242(b)	Compressed Air	● Pneumatic systems must have properly rated components to handle system pressures
29 CFR 1910.95	Occupational Noise Exposure	● Machines must not expose workers to noise levels exceeding 85 decibels (dB) over an 8-hour period without hearing protection. ● Mufflers or dampening mechanisms should be used on pneumatic exhausts to minimize noise.
29 CFR 1910.145	Specifications for Accident Prevention Signs and Tags	● Warning signs must be placed near hazardous areas (e.g., pinch points, emergency stops). ● Signs must use clear language, symbols, and bright colors to ensure visibility.
29 CFR 1910.219	Inspection and Maintenance	● Machines must be inspected regularly to ensure guarding, interlocks, and emergency stops are in good condition. ● Maintenance records must be kept, and all safety devices must be repaired or replaced as needed.

6.0 Logic Analysis

Pneumatics Analysis

The sequence of the cylinders and the pneumatic gripper is as follows:

- CYL-Grip +
- Gripper +
- CYL-Grip -
- CYL-Move +
- CYL-Grip +
- Gripper -
- CYL-Grip -
- CYL-Move -
- CYL-Press +
- CYL-Press -

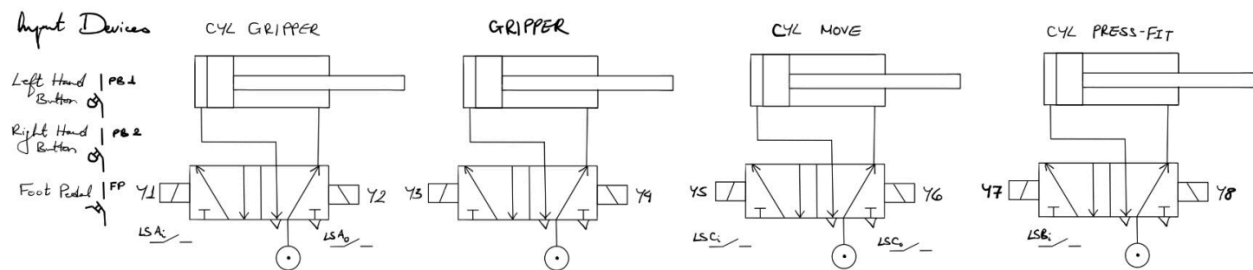


Figure 5. Pneumatics Schematic Depicting Connections

Boolean Analysis

Table 12. Legend for inputs and outputs

Outputs	
CYL-Grip	CYL-A
CYL-Pres	CYL-B
CYL-Move	CYL-C
Inputs	
<i>Push buttons (spring loaded)</i>	
Start button #1	A
Start button #2	B
Foot pedal start	C
<i>Sensors</i>	
Housing detection proximity sensor	D
Limit switch Ai	E
Limit switch Ao	F
Gripper proximity sensor to detect surface its approaching	G
Gripper contact sensor to detect bearing	H
Limit switch Ci	I
Limit switch Co	J
Limit switch Bi	K

Table 13. Truth Table

A	B	C	D	E	F	G	H	I	J	K	OUTPUT
1	1	1	1	1	0	0	0	1	0	1	CYL-A+
1	1	0	1	1	0	0	1	0	1	1	CYL-A+
1	1	0	1	0	1	1	1	1	0	1	CYL-A-
1	1	0	1	0	0	1	0	0	1	1	CYL-A-
1	1	0	1	1	0	0	0	1	0	1	CYL-B+
1	1	0	1	1	0	0	0	0	1	0	CYL-B-
1	1	0	1	1	0	0	1	1	0	1	CYL-C+
1	1	0	1	1	0	0	0	0	1	1	CYL-C-
1	1	0	1	0	1	1	0	1	0	1	Gripper+
1	1	0	1	0	0	1	0	0	1	1	Gripper-

Each row of the truth table (Table 13.) has a different combination of engaged and disengaged sensors which trigger a specific output. Every row must be unique otherwise multiple actions will occur simultaneously which is undesired in this case. This is only a section of the complete truth table showing which combinations were used to create the ladder logic program. The two start push buttons must be continuously engaged by the operator for the cycle to start and continue. These buttons should require both hands to engage to ensure the operator's hands are nowhere near the machine. The foot pedal will be placed under the start push buttons and will be used to initiate the sequence; only momentary engagement is required. The sequence starts with CYL-Grip extending so that the gripper fingers are inside the bearing at which point Li Ai, which will be mounted to the rotating shaft, disengages and Li Ao, which will be mounted to the workstation near the bearing feeder, engages. The proximity sensor mounted in the gripper detects when it is close to the surface it's approaching and the contact sensor, also mounted to the gripper, detects when it is in contact with a bearing. These sensors are required since Li Ao can only be engaged when the shaft is in its initial position, when the shaft is in its second position CYL-Grip must extend into the motor housing where no sensor can be placed without an additional set up. A limit switch (Li Bi) will be mounted with CYL-Press on the rotating shaft to signal when this cylinder is fully retracted. Since CYL-Press must extend into the housing there is no outer limit switch, instead a timer and counter were used in the ladder logic program (Figure 5) to control how long CYL-Press is applying force to the bearing and when it retracts. CYL-Move has an inner and outer limit switch (Li Ci and Li Co) to signal what position this cylinder is in.

7.0 PLC Model

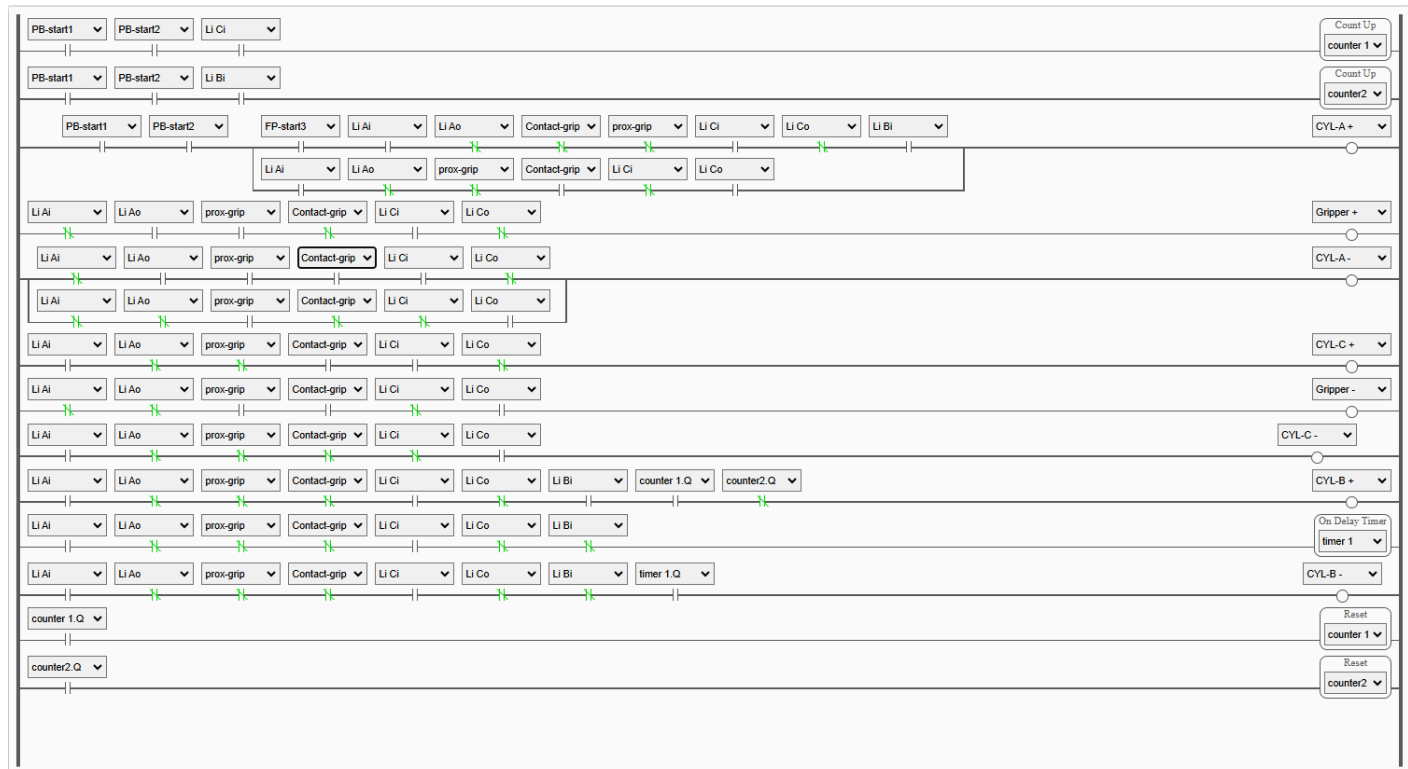


Figure 6. PLC Fiddle Ladder Logic Program

Figure 6 shows the [ladder logic program](#) used to control the circuit. The state of each input will determine the output. Since no sensors could be placed within the motor housing, a timer was used to control CYL-Press and to give it enough time to complete the press fit. The program gives 6 sec for the press fit. Counters were also used to control some movements since CYL-Grip has to extend and retract twice within one cycle.

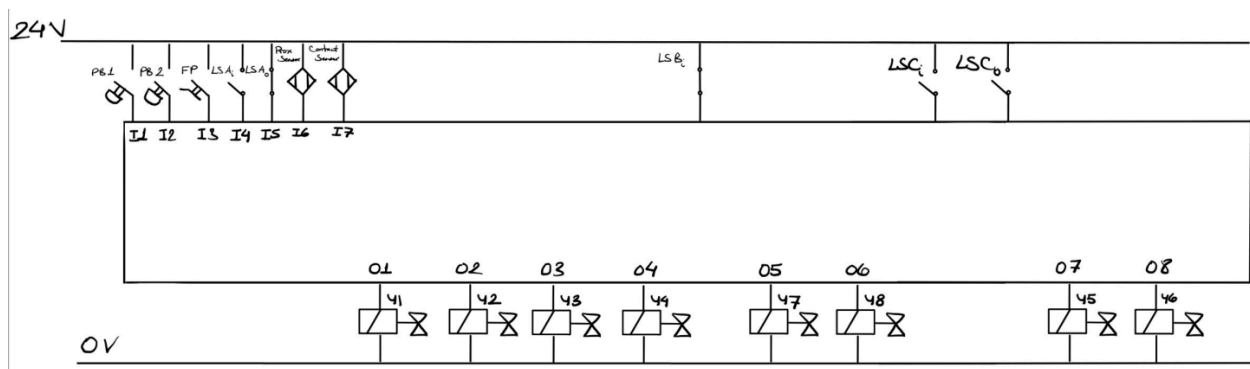


Figure 7: PLC Wiring Diagram

8.0 Conclusions and Recommendations

This project has demonstrated an integration of pneumatic systems with PLC controls. The goals of achieving a cycle time under 30 seconds and ensuring proper bearing alignment have been met. Additionally, the design has adhered to OSHA safety standards to ensure that any workers are able to work this job with the safety precautions set in place. This includes the use of safety enclosures, interlocks, emergency stop features and noise-reduction mufflers. The use of the Festo DSNU pneumatic cylinders is critical to the success of the project because these components were carefully chosen to meet the specific force and stroke length requirements of the system. Moreover, the use of sensors and ladder logic programming ensures smooth operation and accurate feedback throughout the process. However some challenges such as achieving precise angular movement in the rotating assembly remain persistent because of the complexity of coordinating multiple pneumatic cylinders and maintaining consistent force and alignment throughout the rotation. Slight inconsistencies in extension or retraction speeds in these cylinders can introduce errors in the angular positioning.

With hindsight, and if given more time, improvements such as refining the design of the rotating assembly to reduce mechanical complexity and enhance precision could be implemented. Also, the gripper design could also be enhanced to ensure better handling of the bearing while minimizing the risk of damage to its inner race.

9.0 References

1. Festo Catalogue. Accessed: Nov. 29, 2024. [Online]. Available: https://www.festo.com/ca/en/c/products/actuators-and-drives/pneumatic-cylinders/piston-rod-cylinders-id_pim215/
2. Wevolver, "Pick-and-Place Robots: An In-Depth Guide to Their Functionality and Applications." Accessed: Nov. 29, 2024. [Online]. Available: <https://www.wevolver.com/article/pick-and-place-robots-an-in-depth-guide-to-their-functionality-and-applications>
3. Tronstol, "The Three Major Development Trends of the Pick and Place Machine in the Future." Accessed: Nov. 29, 2024. [Online]. Available: <https://www.tronstol.com/the-three-major-development-trends-of-the-pick-and-place-machine-in-the-future>
4. Electronicsforu, "Rise of Advanced Pick & Place Technology in Modern Manufacturing." Accessed: Nov. 29, 2024. [Online]. Available: <https://www.electronicsforu.com/technology-trends/rise-of-advanced-pick-place-technology-in-modern-manufacturing>
5. OSHA, "Laws & Regulations." Accessed: Nov. 29, 2024. [Online]. Available: <https://www.osha.gov/laws-regs>
6. RealPars, "From Wiring Diagram to PLC Program." Accessed: Nov. 29, 2024. [Online]. Available: <https://www.realpars.com/blog/wiring-diagram-to-plc-program>
7. ACC Automation, "Wiring an Inductive Proximity NPN / PNP Sensor to the Click PLC." Accessed: Nov. 29, 2024. [Online]. Available: <https://accautomation.ca/wiring-an-inductive-proximity-npn-pnp-sensor-to-the-click-plc/>
8. Grainger, bearings Accessed: Nov. 29, 2024. [Online]Available: https://www.grainger.ca/en/product/p/WWG36UY36?cm_mmc=PPC:+Google+PLA&gclid=N:N:PS:Paid:GGL:CSM-8500:NOYSQ4:20501231:APZ_1&gclsrc=aw.ds&gad_source=1&gclid=Cj0KCQiAgJa6BhCOARIsAMiL7V8b8US4ZijozZwrCkK0HvxiAyg2jO0gJygTfcT0_DklJzbEdktteS4aAkA_EALw_wcB
9. W. D. Callister and D. G. Rethwisch, Materials science and engineering: An introduction, 9th edition.
10. Engineers Edge, "Coefficients of Friction," Engineers Edge. [Online]. Available: https://www.engineersedge.com/coefficients_of_friction.htm. [Accessed: Nov. 29, 2024].

10.0 Appendix

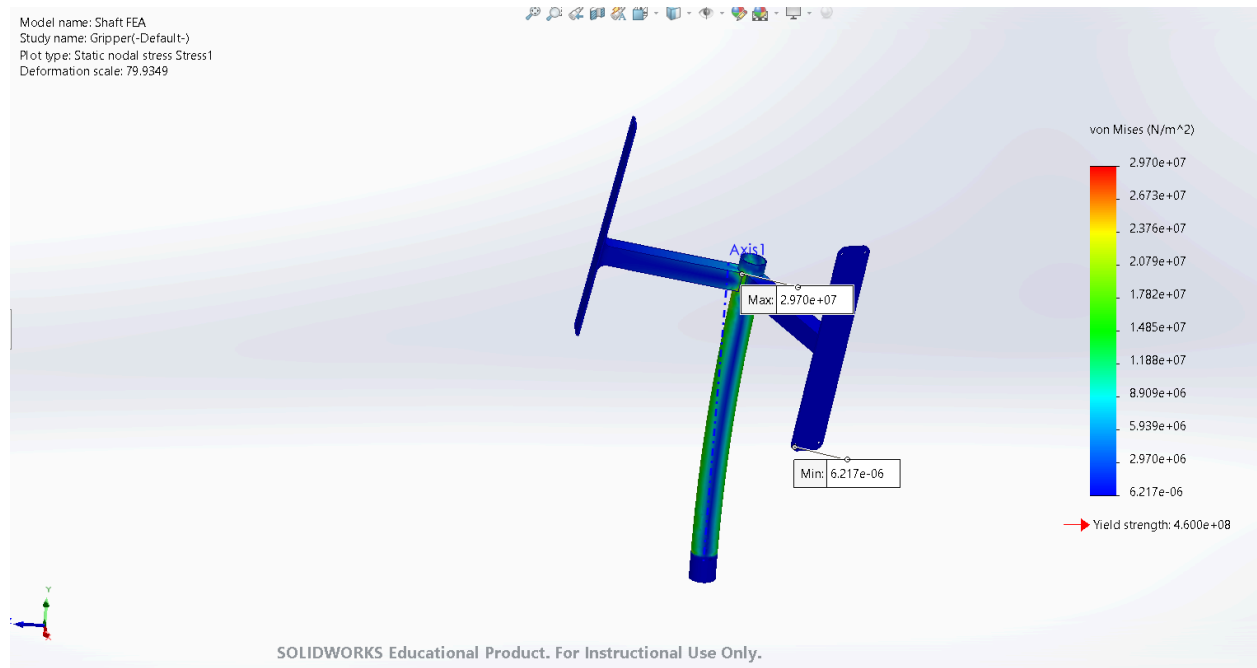


Figure 8: Von Mises Stress of Main Shaft when Gripper cylinder extends

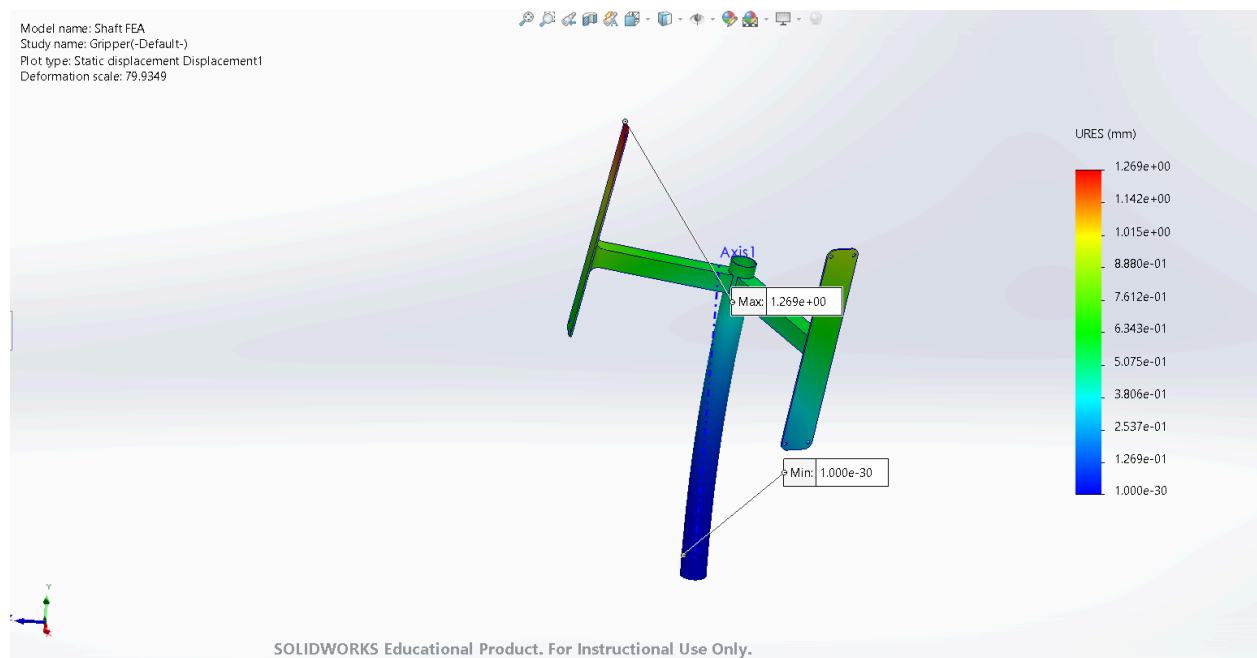


Figure 9: Displacement Main Shaft when Gripper cylinder extends



Figure 10: Factor of Safety Main Shaft when Gripper cylinder extends

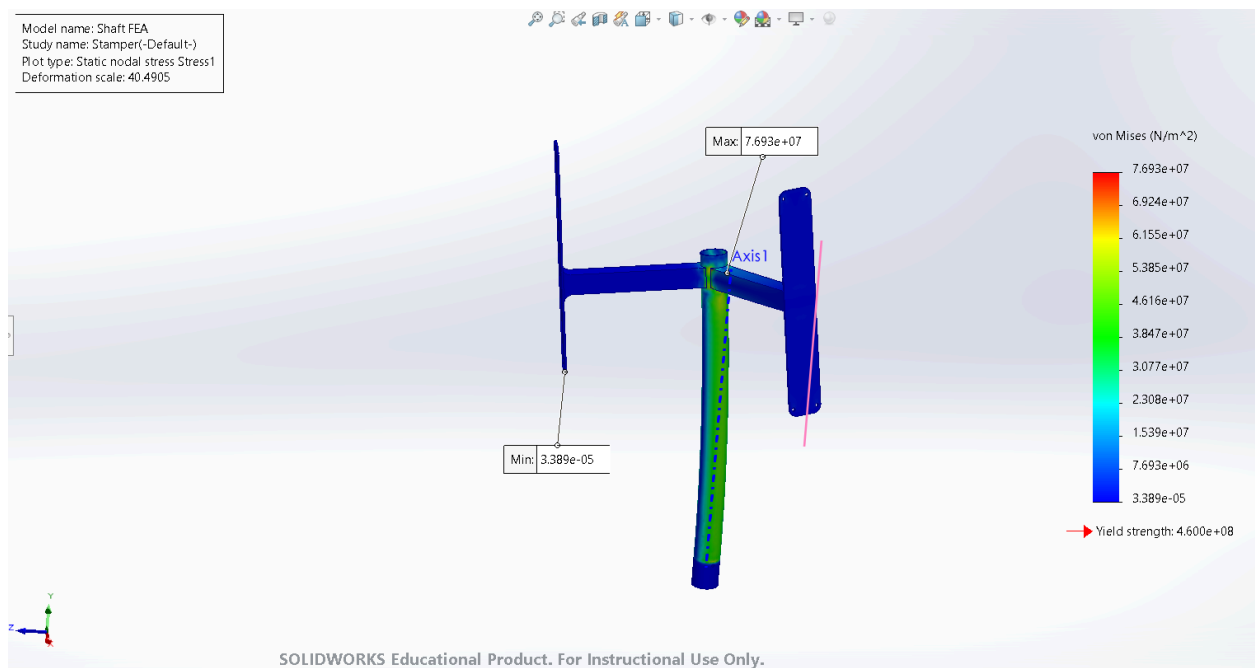


Figure 11: Von Mises Stress of Main Shaft when Press cylinder extends

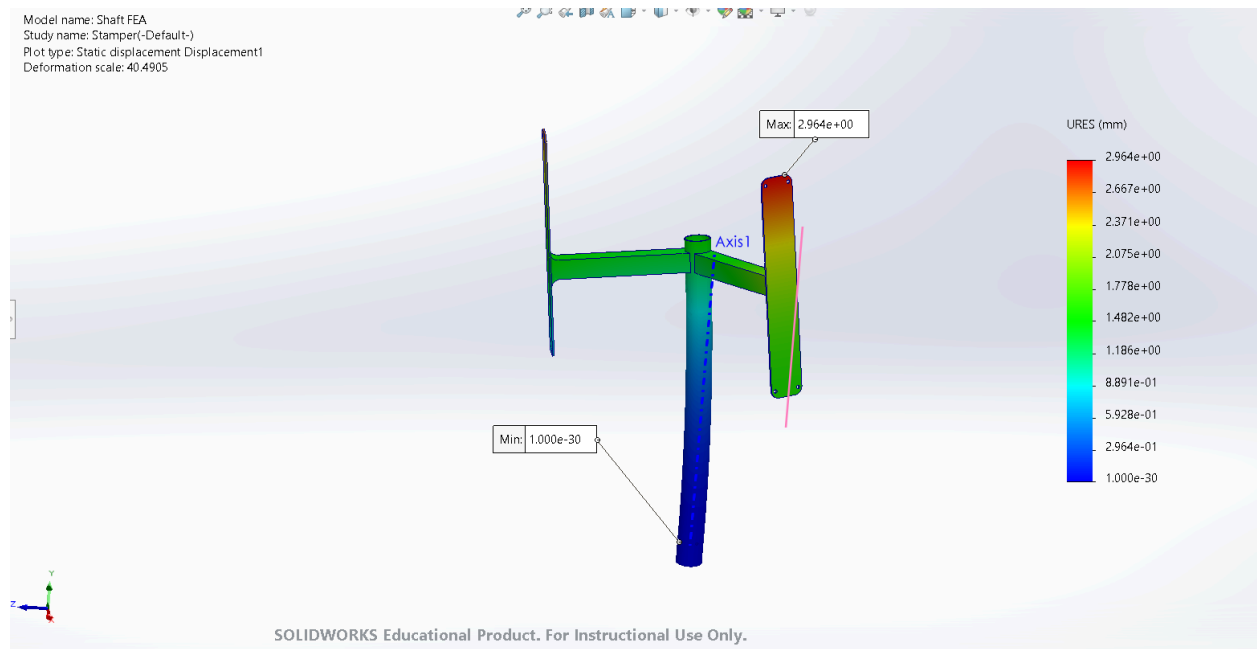


Figure 12: Displacement of Main Shaft when Press cylinder extends

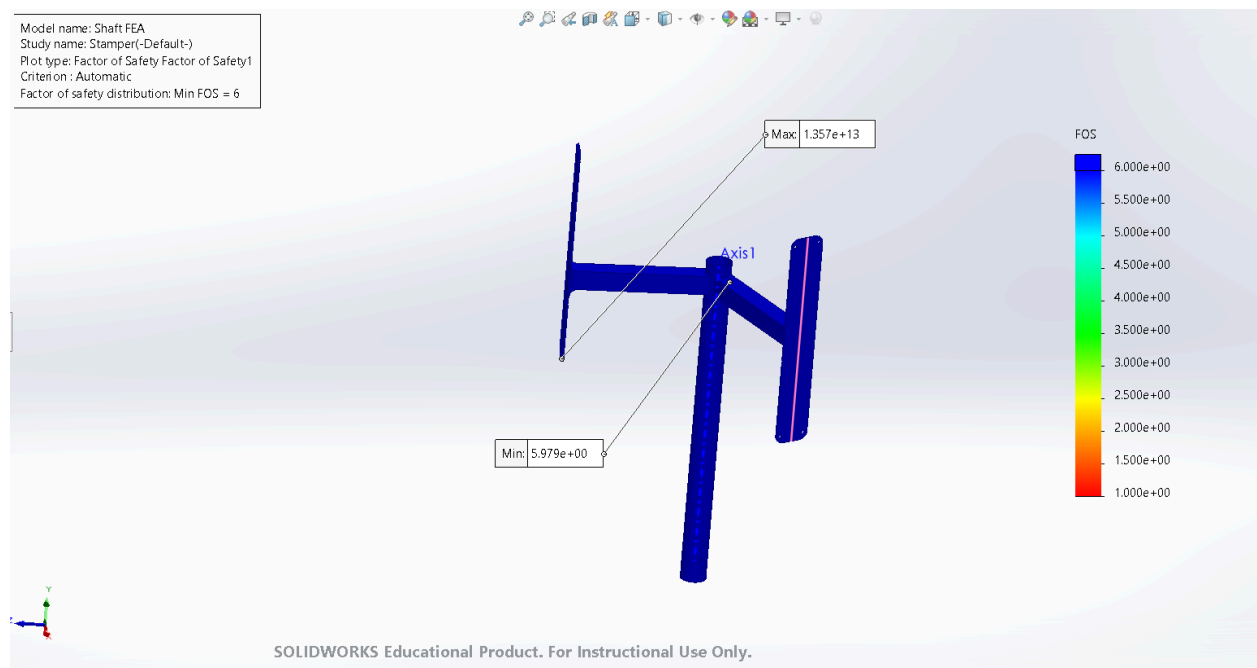


Figure 13: Factor of Safety of Main Shaft when Press cylinder extends

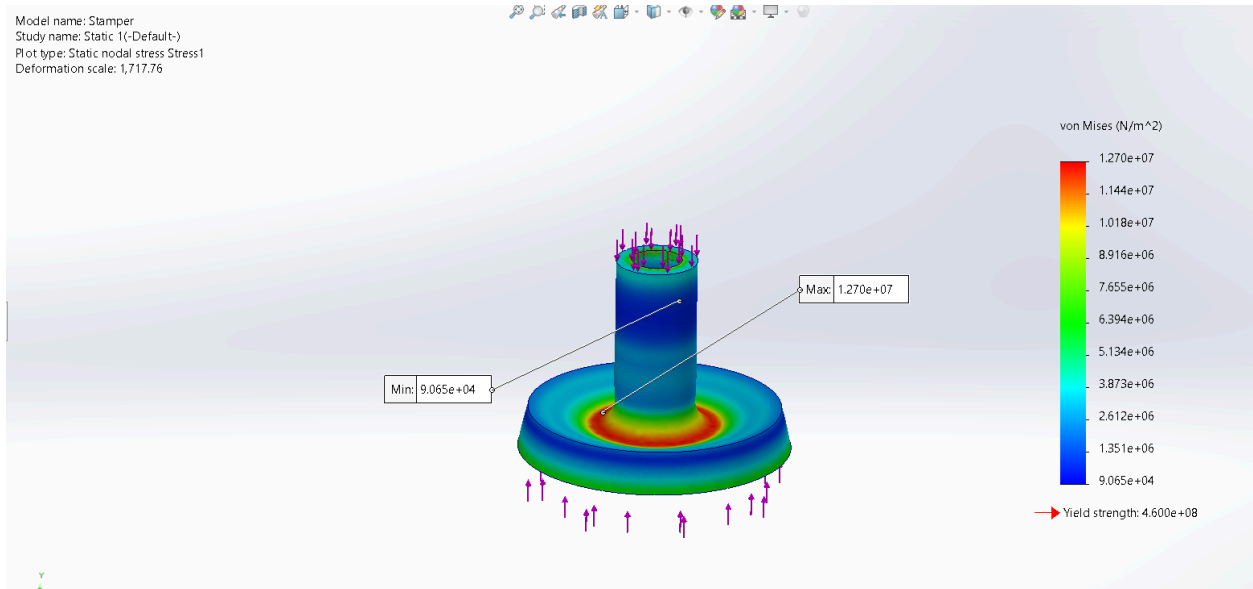


Figure 14: Von Mises Stress of Bearing Stamper

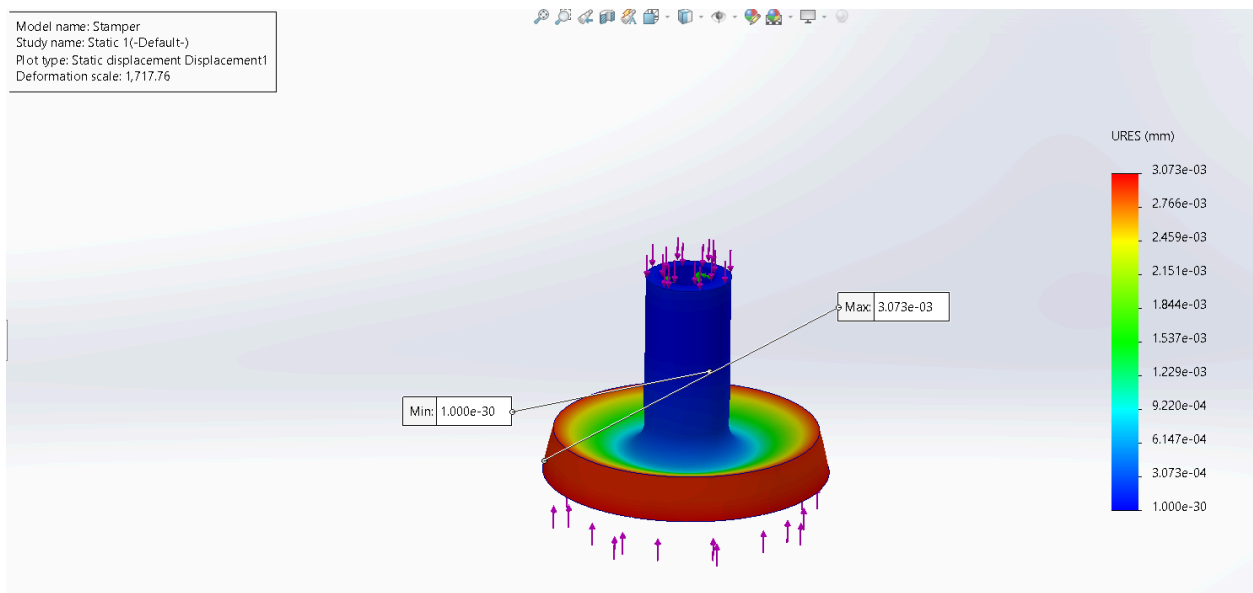


Figure 15: Deformation of Bearing Stamper

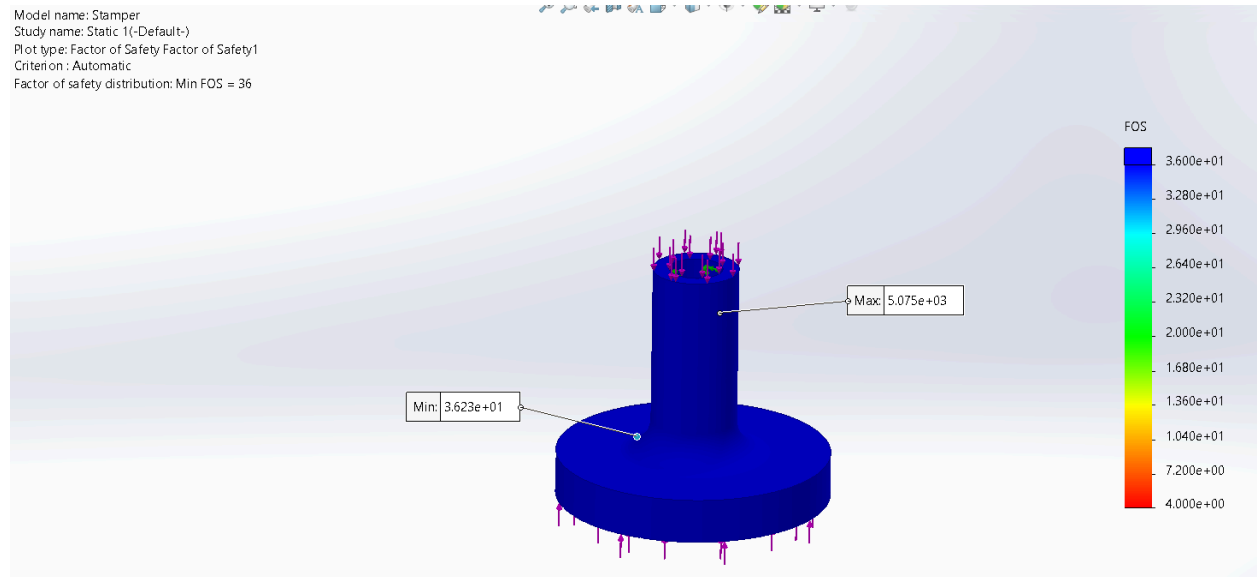


Figure 16: Factor of Safety of Bearing Stamper

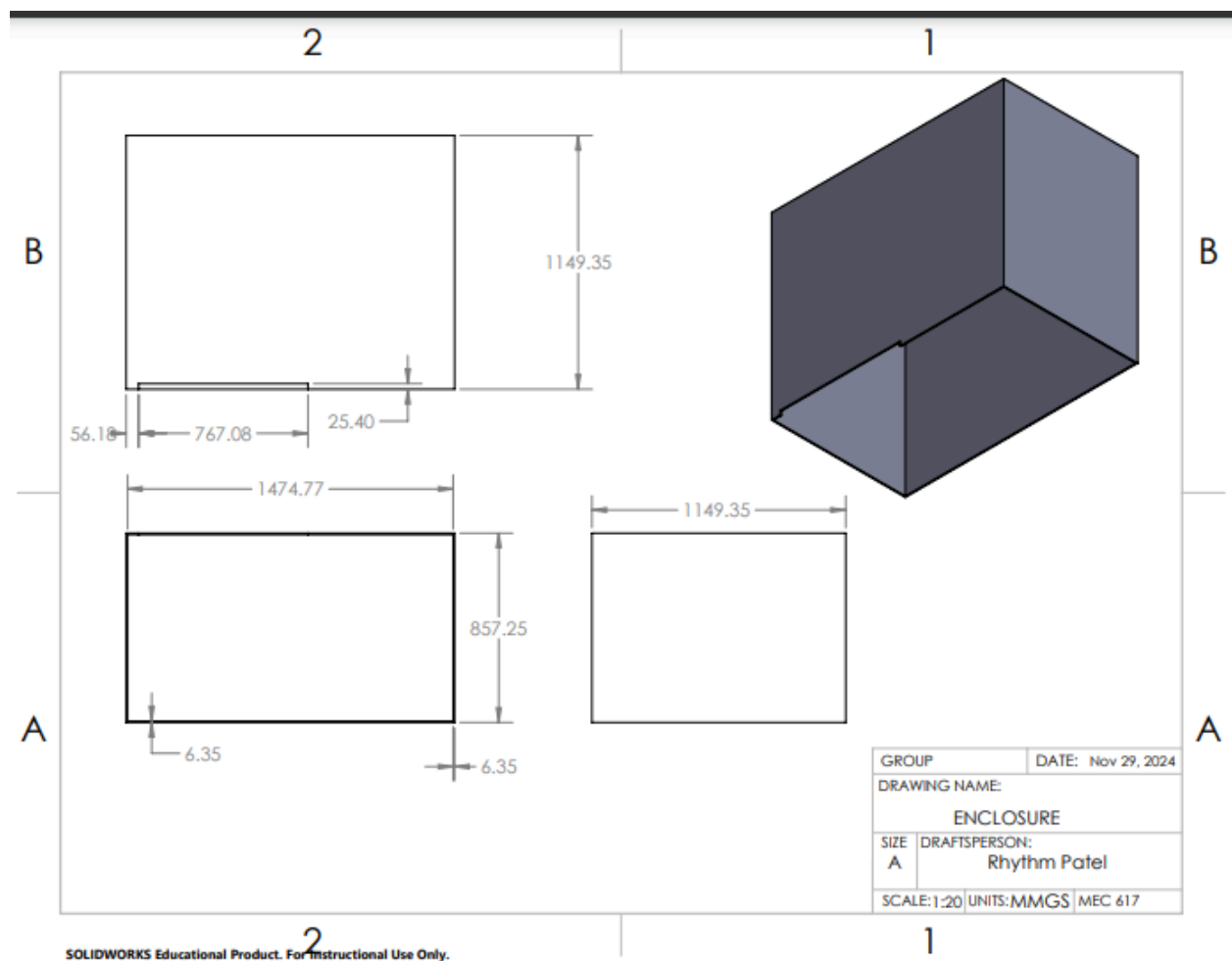


Figure 17: Drawing of Enclosure

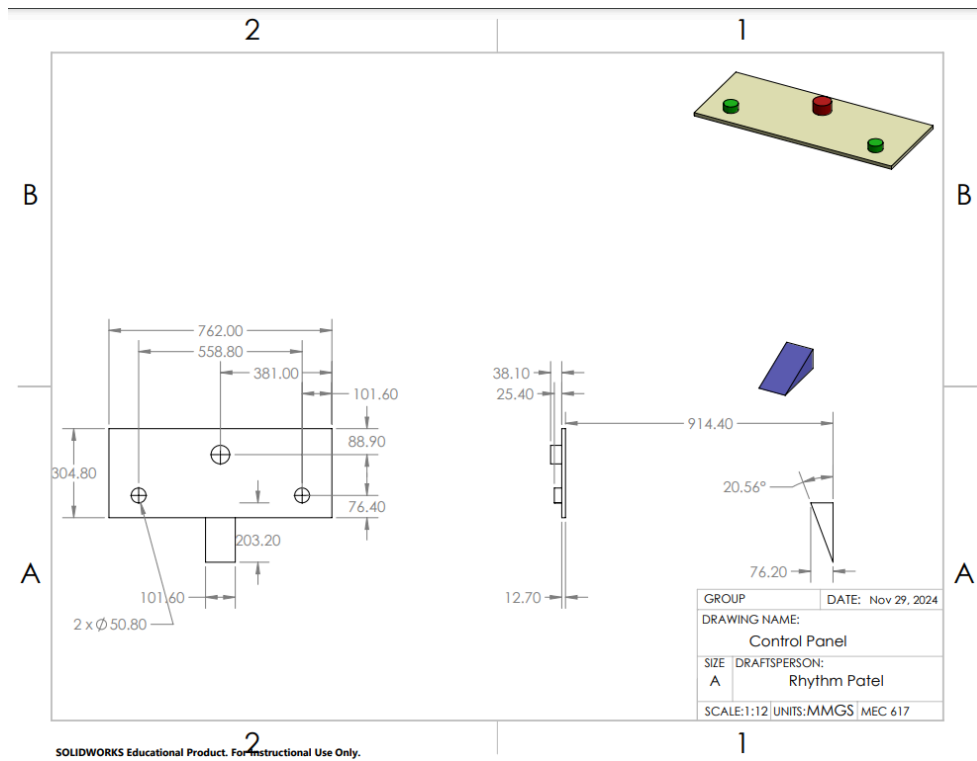


Figure 18: Control Panel Drawing

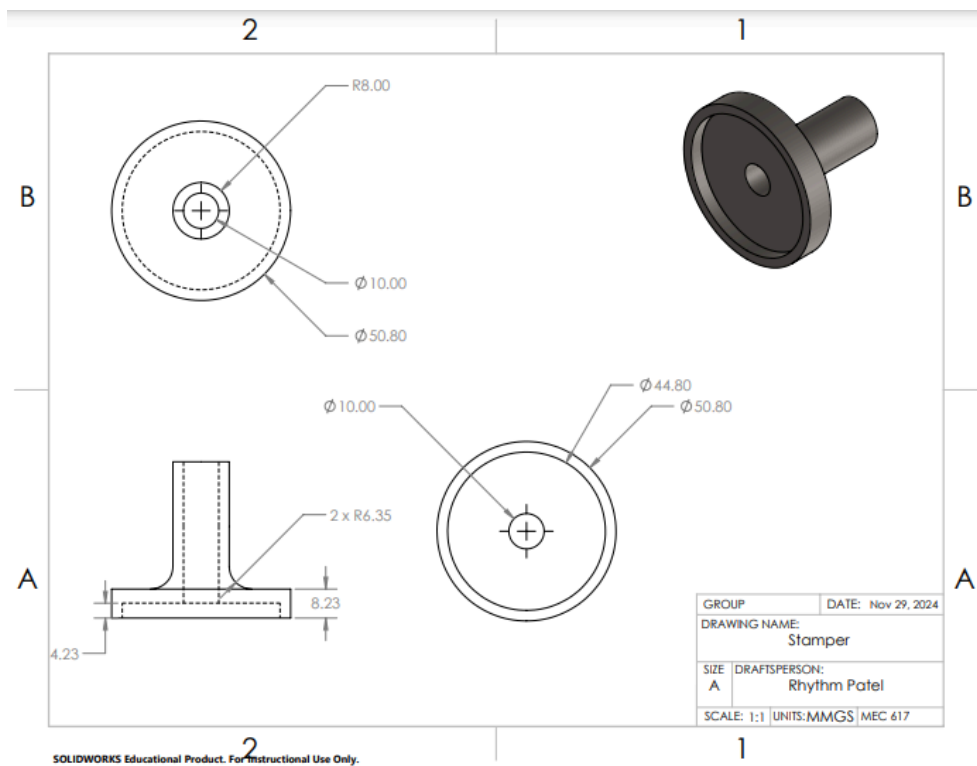


Figure 19: Stamper Drawing

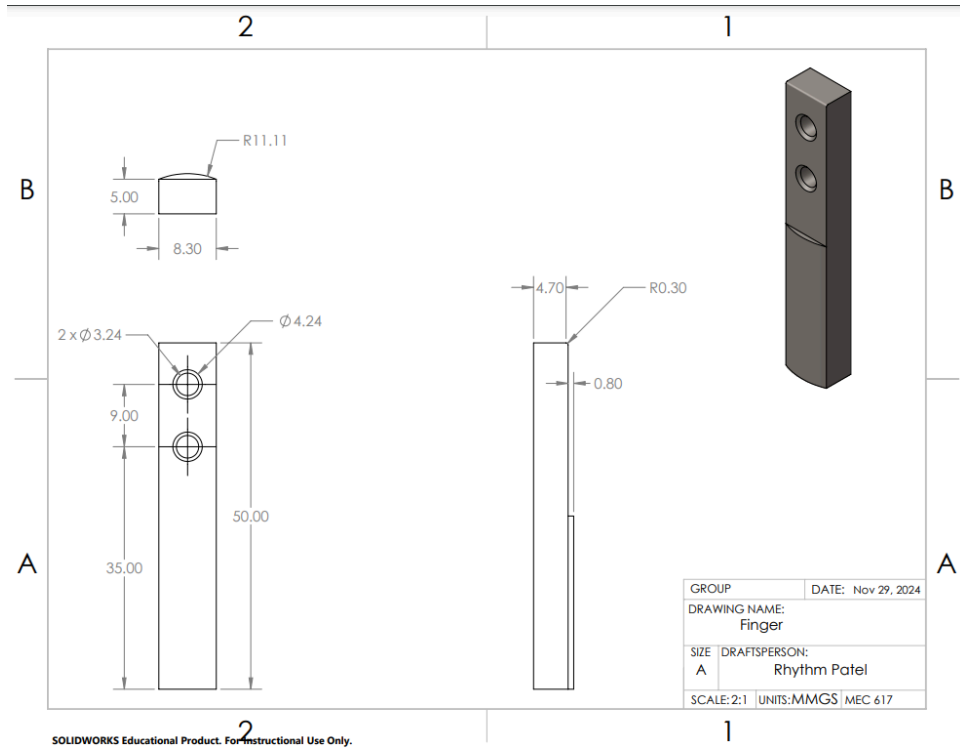


Figure 20: Finger Drawing

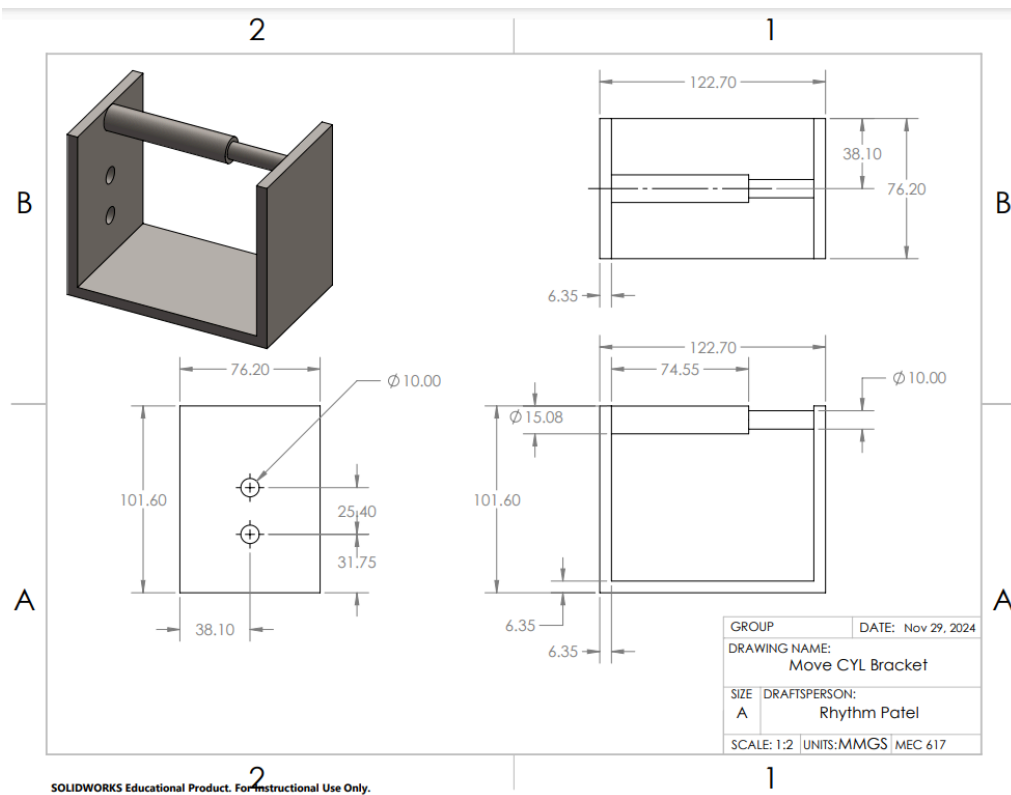


Figure 21: Move Cylinder Bracket Drawing

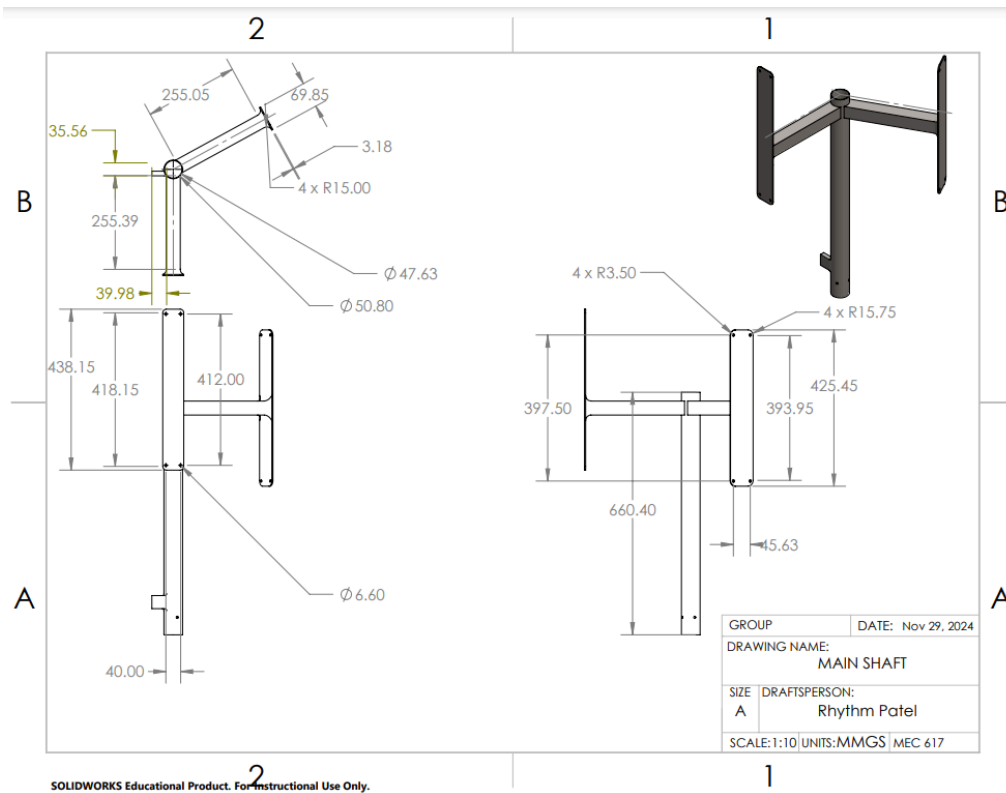


Figure 22: Main Shaft Drawing

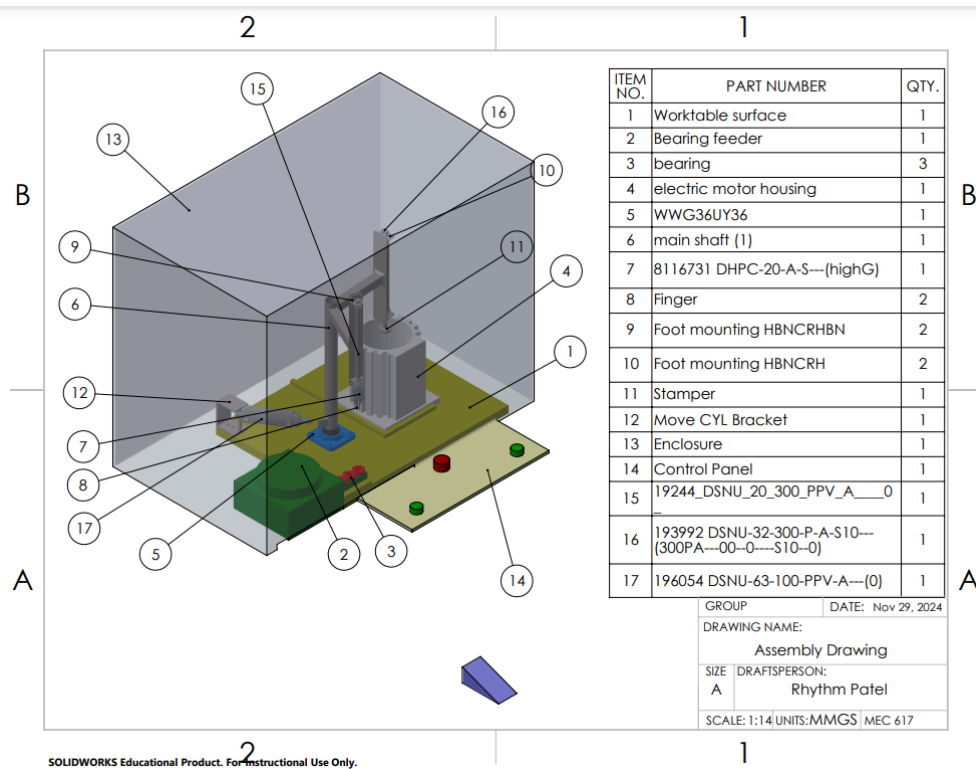


Figure 23: Assembly Drawing